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$$K_C = \frac{[H^+(aq)][HCO_3^-(aq)]}{[CO_2(g)]}$$

Example

Write the K_p and K_c for the following reactions

1) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

2) $2CO(g) \rightleftharpoons CO_2(g) + C(S)$

3) $Ag_2O(S) + 2NH_3(aq) \rightleftharpoons 2AgNO_3(aq) + H_2O(l)$

1) $K_C = \frac{[SO_3]^2}{[SO_2]^2[O_2]}$

$K_P = \frac{P^2_{SO_3}}{P^2_{SO_2} \times PO_2}$

2) $\therefore K_C = \frac{[CO_2]}{[CO]^2}$ and $K_P = \frac{P_{CO_2}}{P^2_{CO}}$

3) $K_C = \frac{[AgNO_3]^2}{[NH_3]^2}$

Evaluate Yourself

1) Consider the following reaction

$$Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons [Fe(SCN)]^{2+}(aq)$$

A solution is made with initial Fe^{3+}, SCN^- concentration of 1×10^{-3}M and 8×10^{-4} M respectively. At equilibrium $[Fe(SCN)]^{2+}$ concentration is 2×10^{-4}M. Calculate the value of equilibrium constant.

2) The atmospheric oxidation of NO

$$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$$

was studied with initial pressure of 1 atm of NO and 1 atm of O_2. At equilibrium, partial pressure of oxygen is 0.52 atm calculate K_p of the reaction.